

Learning Material

For the purposes of the examination and project artefact, the learning material will **not cover linear polynomials or any polynomials higher than 3th degree polynomials** (i.e. quartic and quintic polynomials). The quartic formula is too unwieldy to be used within this context – the image below displays only one solution to the general quartic formula.

Figure 1 – AoPSOnline (n.d.)

Quartic polynomial equation general formula

$$z = \frac{b}{4a} + \frac{1}{2} \sqrt{\frac{3b^2 - 8ac}{12a^2} + \frac{1}{3a} \left(\frac{-3c^2 - 9bcd + 27b^2e + 27ad^2 - 72ace + \sqrt{(2c^2 - 9bcd + 27b^2e + 27ad^2 - 72ace)^2 - 4(c^2 - 3bd + 12ae)^2}}{2} + \frac{c^2 - 3bd + 12ae}{\sqrt{(2c^2 - 9bcd + 27b^2e + 27ad^2 - 72ace)^2 - 4(c^2 - 3bd + 12ae)^2}} \right)} \pm \frac{1}{2} \sqrt{-4 \left(\frac{3b^2 - 8ac}{12a^2} + \frac{1}{3a} \left(\frac{-3c^2 - 9bcd + 27b^2e + 27ad^2 - 72ace + \sqrt{(2c^2 - 9bcd + 27b^2e + 27ad^2 - 72ace)^2 - 4(c^2 - 3bd + 12ae)^2}}{2} + \frac{c^2 - 3bd + 12ae}{\sqrt{(2c^2 - 9bcd + 27b^2e + 27ad^2 - 72ace)^2 - 4(c^2 - 3bd + 12ae)^2}} \right) \right) + \frac{8ac - 3b^2}{4a^2} + \frac{\frac{b^3 - 4abc + 6ad^2}{3a^2}}{\sqrt{(2c^2 - 9bcd + 27b^2e + 27ad^2 - 72ace)^2 - 4(c^2 - 3bd + 12ae)^2}} + \frac{c^2 - 3bd + 12ae}{\sqrt{(2c^2 - 9bcd + 27b^2e + 27ad^2 - 72ace)^2 - 4(c^2 - 3bd + 12ae)^2}}}$$

The quintic formula and higher degree polynomials **do not exist** in the sense that **there's no general formula** using only addition, subtraction, multiplication, division, and nth roots (radicals) to solve all quintic polynomials—polynomials of degree 5, like $ax^5 + bx^4 + cx^3 + dx^2 + ex + f = 0$ —for their roots.

What are polynomials?

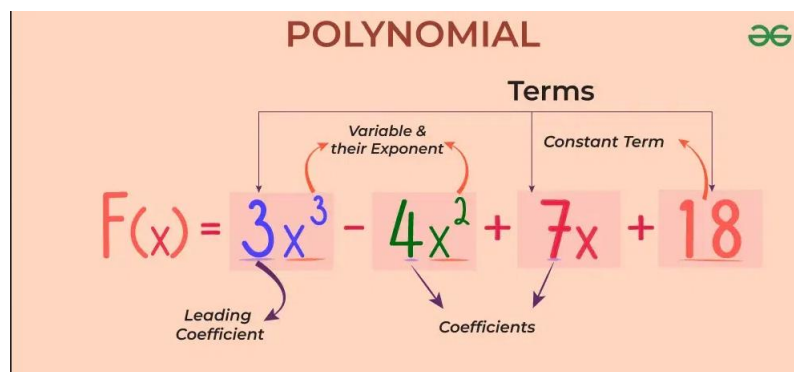
A polynomial is a **mathematical expression made up of terms** that are combined using addition, subtraction, and multiplication. Each term consists of a variable (like x or y) raised to a non-negative integer power, multiplied by a coefficient (a number). For example, **$3x^2 + 5x - 2$ is a polynomial**. The highest power of the variable in the polynomial determines its degree—in this case, the degree is 2 because of the x^2 term. Polynomials can have one variable (univariate) or multiple variables (multivariate), and

they're widely used in algebra to model relationships and solve equations (greatly contributed from the Fundamental theorem of Algebra).

The **discriminant of a polynomial** is a function of the coefficients of the polynomial. It is helpful in determining what type of solutions a polynomial equation has without finding them.

Figure II – GeeksForGeeks (n.d.)

Polynomial structure



The Fundamental theorem of Algebra

The *fundamental theorem of algebra* states that all **single-variable polynomial of degree n will have exactly N roots that exist within the real or complex plane.**

If you have a polynomial (like $x^2 + 1$ or $x^3 - 2x + 3$) with complex number coefficients, it will always have at least one solution (or root) in the complex number system. This also implies that a polynomial of degree n will have exactly n roots in the complex plane, counting multiplicities (i.e., some roots might be repeated). For example, a quadratic equation (degree 2) has 2 roots, a cubic (degree 3) has 3 roots, and so on. **This theorem guarantees solutions exist, even if they're not real numbers.**

What are Complex numbers?

Complex numbers are numbers that have two parts: **a real part and an imaginary part.**

They're written in the form $a + bi$, where (a) is the real part, (b) is the imaginary part, and (i) is the imaginary unit, defined by the property:

$$i^2 = -1 \text{ or } i = \sqrt{-1}$$

$3 + 4i$ is a complex number with real part 3 and imaginary part 4. They were introduced to solve equations like $x^2 + 1 = 0$, which have no solutions among real numbers (since squaring any real number gives a non-negative result). In the complex plane, they're represented as points, with the real part on the x-axis and the imaginary part on the y-axis. Complex numbers are crucial in fields like engineering, physics, and math; especially when dealing with polynomials, where every non-constant polynomial has roots.

What is a complex conjugate?

A **complex conjugate** of a complex number is a number that **flips the sign of the imaginary part** while keeping the real part the same. If you have a complex number

$$z = a + bi$$

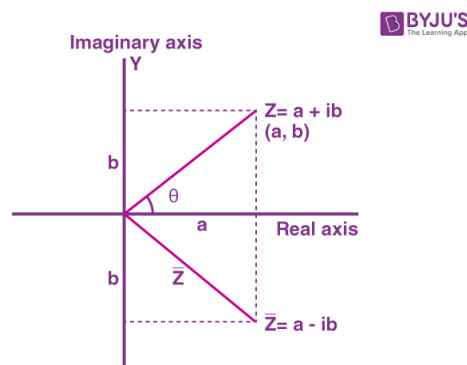
(where (a) is the real part and (b) is the imaginary part), its complex conjugate is

$$z^* = a - bi$$

In this situation, the complex conjugate of the complex number $z = (3 + 4i)$ is $(3 - 4i)$.

Figure III – Byju (n.d.)

Complex Conjugate graphical representation



Complex number operations

You can add, subtract, multiply, and divide them:

Addition & Subtraction:

Figure IV – CueMath (n.d.)

Addition & Subtraction of Complex numbers

Adding and Subtracting Complex Numbers



$$(a + ib) + (c + id) = (a + c) + i(b + d)$$

$$(a + ib) - (c + id) = (a - c) + i(b - d)$$

Multiplication:

Use the distributive property and substitute ($i^2 = -1$)

Figure V – CueMath (n.d.)

Multiplication of Complex numbers

Multiplying Complex Numbers Formula



$$(a + ib)(c + id) = (ac - bd) + i(ad + bc)$$

Division:

To divide, multiply the numerator and denominator by the conjugate of the denominator.

$$\frac{(1+i)}{(1-i)} = \frac{(1+i)(1+i)}{(1-i)(1+i)} = \frac{1+2i+i^2}{1-i^2} = \frac{1+2i+i^2}{1-i^2} = \frac{1+2i-1}{2} = \frac{2i}{2} = i$$

Figure VI – CueMath (n.d.)

Division of Complex numbers

$$\begin{aligned}
 \frac{a + bi}{c + di} &= \frac{(a + bi)(c - di)}{(c + di)(c - di)} \\
 &= \frac{(ac + bd) + (bc - ad)i}{c^2 + d^2} \\
 &= \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2} i
 \end{aligned}$$

Polynomial Varieties

Quadratic polynomial

Quadratic polynomials **are polynomials of degree 2**, meaning the highest power of the variable is 2. They can be written in the form:

$[f(x) = ax^2 + bx + c]$ where: a, b, and c are constants (numbers), and $a \neq 0$

The graph of a quadratic polynomial is a **parabola**, which opens upward if $a > 0$ or downward if $a < 0$. Thanks to the Fundamental Theorem of Algebra, a **quadratic polynomial has exactly 2 roots** in the complex numbers. You can find these roots using the quadratic formula

Figure VII – Lupercio (2022)

Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

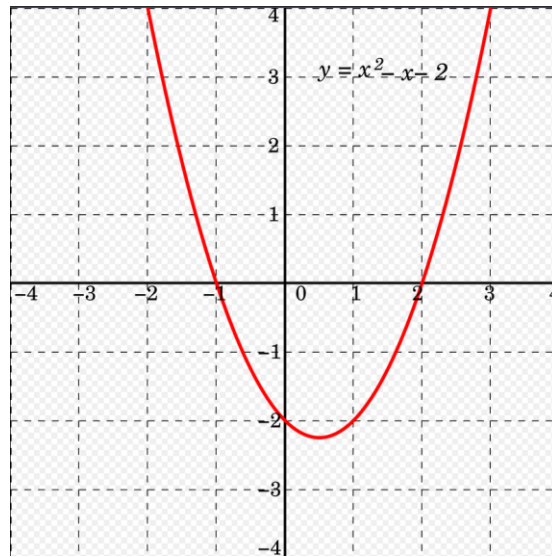
Quadratic discriminant

The formula, $\Delta = \mathbf{b^2 - 4ac}$, determines the nature of the roots:

- If $\Delta > 0$, there are 2 distinct real roots.
- If $\Delta = 0$, there's 1 real root (double root).
- If $\Delta < 0$, there are 2 complex conjugate roots.

Figure VIII – Wikipedia (n.d.)

Graphical representation of a quadratic equation



Cubic polynomial

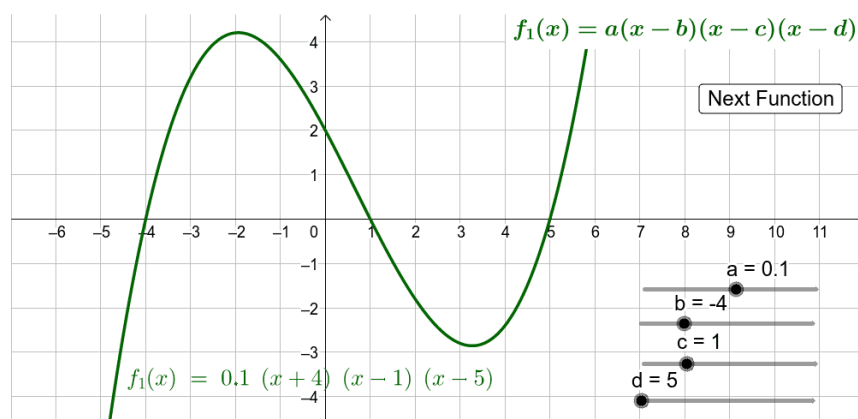
Cubic polynomials **are polynomials of degree 3**, meaning the highest power of the variable is 3. They can be written in the form:

$[f(x) = ax^3 + bx^2 + cx + d]$ where: a, b, c and d are constants (numbers), and $a \neq 0$

The graph of a cubic polynomial is a curve that can have up to two "turns" or bends. It doesn't just sit flat or open like a parabola—it stretches from negative to positive infinity (or vice versa) because odd-degree polynomials always have opposite behaviour at their ends. If $a > 0$, it rises from left to right; if $a < 0$, it falls.

Figure IX – Sassman (2017)

Cubic equation visual representation



Cubic discriminant

For a standard cubic in the form $f(x) = ax^3 + bx^2 + cx + d$, the formula, $\Delta = b^2c^2 - 4ac^3 - 4b^3d - 27a^2d^2 + 18abcd$, determines the nature of the roots:

- If $\Delta > 0$, $f(x)$ has three distinct real roots.
- If $\Delta = 0$, $f(x)$ has a repeated root, and all its roots are real.
- If $\Delta < 0$, $f(x)$ has one real root and two non-real complex conjugate roots.

Cubic equation

Cubic equations can cross the x-axis **up to 3 times** if all roots are real and distinct or fewer if some roots are complex or repeated. They also have at least one real root because odd-degree polynomials always cross the x-axis at least once. Although quite expansive, this is the equation used to solve for cubic polynomials.

Cubic Formula

$$x^1 = -\frac{b}{3a} + \sqrt[3]{\left(\frac{b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right) + \sqrt{\left(\frac{b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right)^2 + \left(\frac{c}{3a} - \frac{b^2}{9a^2}\right)^3}} + \sqrt[3]{\left(\frac{b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right) - \sqrt{\left(\frac{b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right)^2 + \left(\frac{c}{3a} - \frac{b^2}{9a^2}\right)^3}}$$

$$x^2 = -\frac{b}{3a} + \frac{-1 + i\sqrt{3}}{2} \sqrt[3]{\left(\frac{b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right) + \sqrt{\left(\frac{b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right)^2 + \left(\frac{c}{3a} - \frac{b^2}{9a^2}\right)^3}} + \frac{-1 - i\sqrt{3}}{2} \sqrt[3]{\left(\frac{b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right) - \sqrt{\left(\frac{b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right)^2 + \left(\frac{c}{3a} - \frac{b^2}{9a^2}\right)^3}}$$

$$x^3 = -\frac{b}{3a} + \frac{-1 - i\sqrt{3}}{2} \sqrt[3]{\left(\frac{b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right) + \sqrt{\left(\frac{b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right)^2 + \left(\frac{c}{3a} - \frac{b^2}{9a^2}\right)^3}} + \frac{-1 + i\sqrt{3}}{2} \sqrt[3]{\left(\frac{b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right) - \sqrt{\left(\frac{b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right)^2 + \left(\frac{c}{3a} - \frac{b^2}{9a^2}\right)^3}}$$